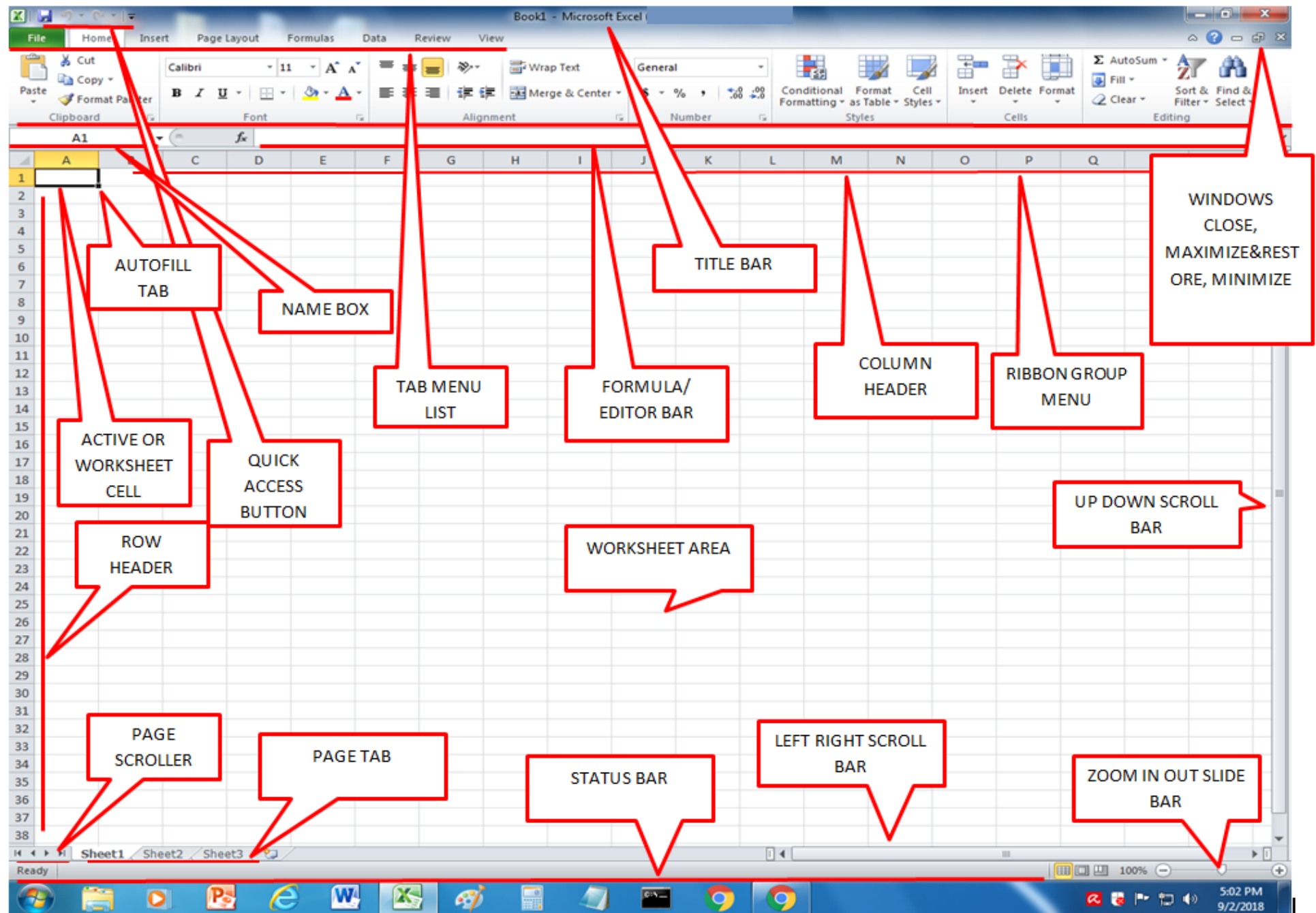


Advanced MSOfficeOpenOffice (Word, Excel, Access, PPT)

Microsoft Excel

What is Microsoft Excel?

- **electronic spreadsheet** - stores numeric data in two-dimensional tables that display the results of calculations performed on these data.
- Developed by Microsoft



MS Excel Window's Elements

- **Quick Access Toolbar** – Collection of buttons that provide one click access to commonly used commands such as Save, Undo or Redo.
- **Title Bar** – A bar that displays the name of the active workbook.
- **Ribbon** – The main set of commands and controls organized into tabs and groups, you can also customize the ribbon according to your preference.
- **Column Headings** – The letters that appear along the top of the worksheet to identify the different columns in the worksheet.

MS Excel Window's Elements

- **Worksheet Window** – A window that displays an Excel worksheet, basically this is where you work all the tasks.
- **Vertical Scroll Bar** – Scroll bar to use when you want to scroll vertically through the worksheet window.
- **Horizontal Scroll Bar** – Scroll bar to use when you want to scroll horizontally through the worksheet window.
- **Zoom Controls** – Used for magnifying and shrinking of the active worksheet.
- **View Shortcuts** – Buttons used to change how the worksheet content is displayed. Normal, Page Layout or Page Break Preview.

MS Excel Window's Elements

- **Sheet Tabs** – Tabs that display the name of the worksheet in the workbook, by default its name sheet 1, sheet 2, etc. You can rename this to any name that best represents your sheet.
- **Sheet Tab Scrolling Buttons** – Buttons to scroll the sheet tabs in the workbook
- **Row Headings** – The number that appears on the left of the worksheet window to identify the different rows.
- **Select All Button** – A button that selects all the cells in the active worksheet
- **Active Cell** – The cell selected in the active worksheet
- **Name Box** – A box that displays the cell reference of the active cell

MS Excel Window's Elements

- **Formula Bar** – A bar that displays the value or formula entered in the active cell
- **File Tab** – It provides access to workbook level features and program settings.

Workbooks and worksheets

- A **workbook** automatically shows in the workspace when you open Microsoft Excel XP. Each workbook contains three **worksheets**. A worksheet is a grid of cells consisting of 65,536 rows by 256 columns. Spreadsheet information—text, numbers, or mathematical formulas—is entered into different cells.

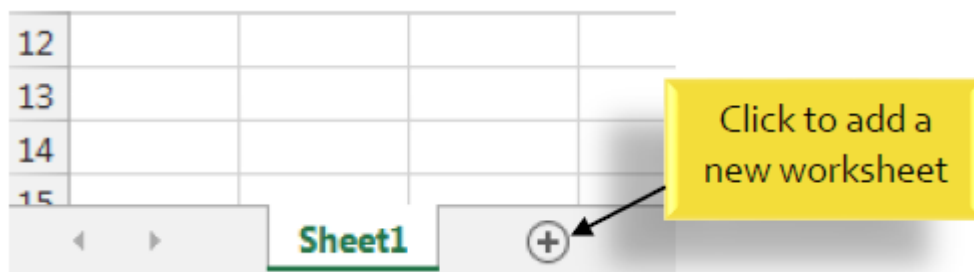
MS Excel Window's Elements

- **Column headings** are referenced by alphabetic characters in the gray boxes that run across the Excel screen, beginning with column A and ending with column IV.
- **Rows** are referenced by numbers that appear on the left and then run down the Excel screen. The first row is named row 1, while the last row is named 65536.
- **Cell:** intersection of a column and a row.
- **Active cell**—or the cell that can be acted upon—reveals a dark border. All other cells reveal a light gray border. Each cell has a name. Its name is comprised of two parts: the column letter and the row number.

Working with workbooks

To create a new blank workbook

- Select the **File** tab
- Select **New**, then click **Blank workbook**.
- Or
- Press CTRL+N



To insert a new worksheet

- Locate and select the New sheet button.
- A new, blank worksheet will appear.

Working with workbooks

To delete a worksheet

- Right-click the worksheet you wish to delete from the worksheet menu
- Or
- From the Home Tab in the Cells Group click on Delete and select Delete Sheet.
- The worksheet will be deleted from your workbook.

To copy or move a worksheet

- Right-click the worksheet you want to copy, then select Move or Copy from the worksheet menu.
- The Move or Copy dialog box will appear.

Working with workbooks

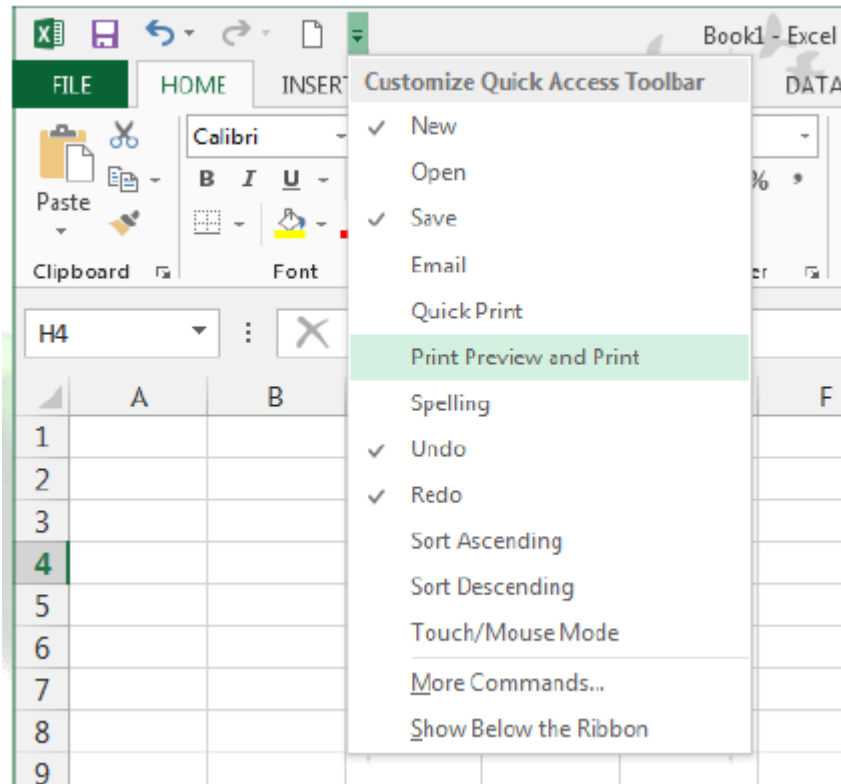
- Choose where the sheet will appear in the Before sheet: field. In our example, we'll choose (move to end) to place the worksheet to the right of the existing worksheet.
- Check the box next to Create a copy, then click OK.
- The worksheet will be copied. It will have the same title as the original worksheet, as well as a version number.
- NB: You can also copy a worksheet to an entirely different workbook. You can select any workbook that is currently open from the To book: drop-down menu.

To add commands to the Quick Access toolbar

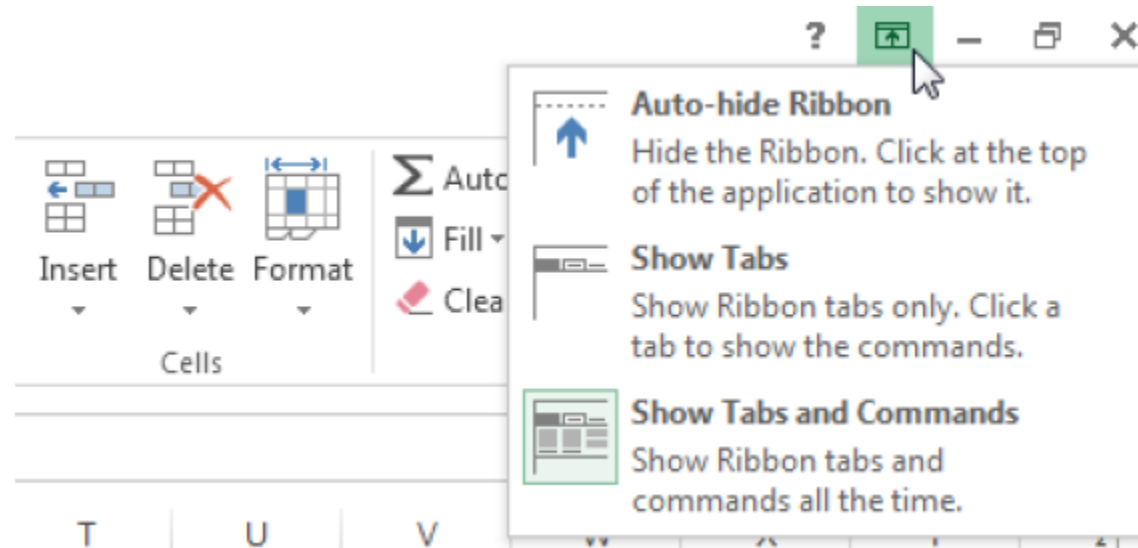
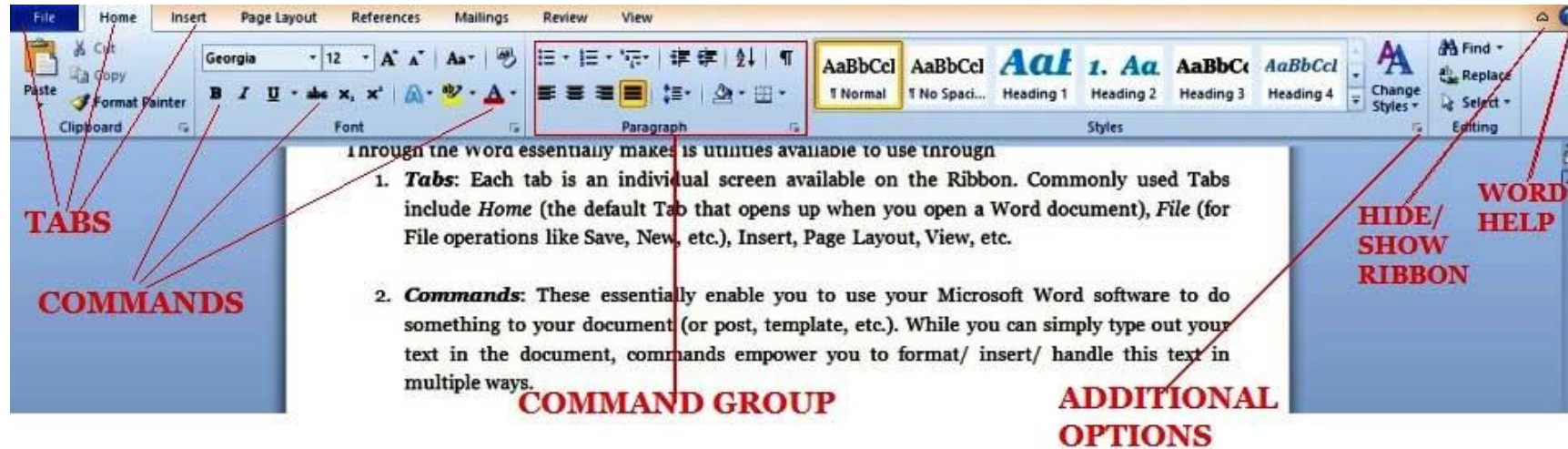
- Click the drop-down arrow to the right of the Quick Access toolbar.

Working with workbooks

- Select the command you wish to add from the drop-down menu. To choose from more commands, select More Commands.
- The command will be added to the Quick Access toolbar.



Ribbon



Ribbon

- Excel uses a tabbed Ribbon system instead of traditional menus. The Ribbon contains multiple tabs, each with several groups of commands. You will use these tabs to perform the most common tasks in Excel.

To minimize and maximize the Ribbon

- The Ribbon is designed to respond to your current task, but you can choose to minimize it if you find that it takes up too much screen space.
- Click the Ribbon Display Options arrow in the upper-right corner of the Ribbon.
- Select the desired minimizing option from the drop-down menu.

Ribbon

- **Auto-hide Ribbon:** Auto-hide displays your workbook in full-screen mode and completely hides the Ribbon. To show the Ribbon, click the Expand Ribbon command at the top of screen.
- **Show Tabs:** This option hides all command groups when not in use, but tabs will remain visible. To show the Ribbon, simply click a tab.
- **Show Tabs and Commands:** This option maximizes the Ribbon. All of the tabs and commands will be visible. This option is selected by default when you open Excel for the first time.

Navigating a Worksheet

- Excel provides several ways to navigate a worksheet

Figure 1-5

Excel navigation keys

Press	To move the active cell
↑, ↓, ←, →	Up, down, left, or right one cell
Home	To column A of the current row
Ctrl+Home	To cell A1
Ctrl+End	To the last cell in the worksheet that contains data
Enter	Down one row or to the start of the next row of data
Shift+Enter	Up one row
Tab	One column to the right
Shift+Tab	One column to the left
Page Up, Page Down	Up or down one screen
Ctrl+Page Up, Ctrl+Page Down	To the previous or next sheet in the workbook

Data Entry

- Place your mouse in a cell and click once. This will allow you to enter data in that cell.
- To move HORIZONTALLY across cells, hit TAB.
- To move VERTICALLY, hit ENTER.

Formula

- A formula is a mathematical expression that calculates a value.
- In Excel, formulas always begin with an equal sign (=).
- A formula can consist of one or more arithmetic operators.
- The order of precedence is a set of predefined rules that Excel follows to calculate a formula.

Copying and Pasting Formulas

- With formulas, however, Excel adjusts the formula's cell references to reflect the new location of the formula in the worksheet

Formula copied and pasted

Figure 1-25

formula copied from this cell

formula pasted into cells H8 and H9

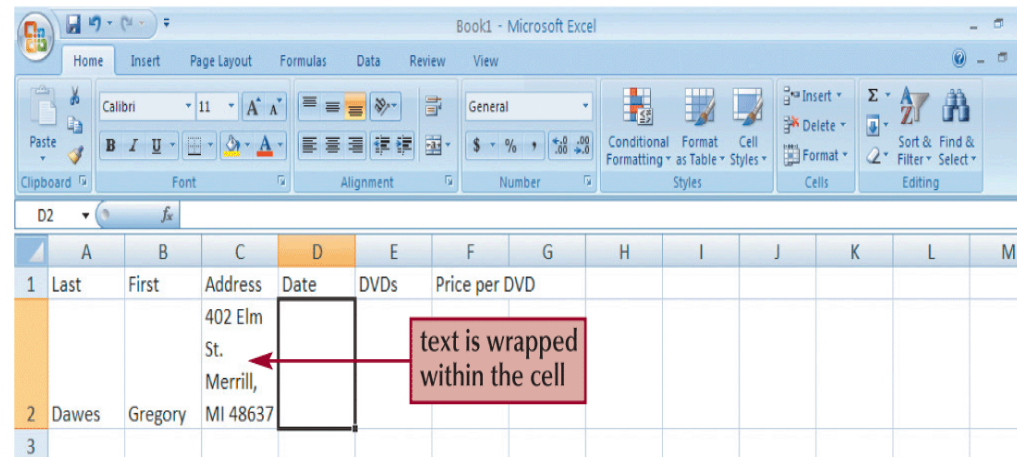
	A	B	C	D	E	F	G	H	I
1	RipCity Digital								
2	Customer Orders								
3	3/31/2010								
4									
5	Last	First	Address	Phone	Date	DVDs	Price per DVD	Charge	
6	Ferris	Andrew	135 College Ave. Bar Harbor, ME 04609	(207) 555-0101	3/5/2010	2	\$17.29	\$34.58	
7	Garcia	Susan	1025 Drake Ave. Exeter, NH 03833	(603) 555-1091	3/14/2010	25	\$15.79	\$394.75	
8	Torbet	Dr. Lilla	5 North Ln. Oswego, NY 13126	(315) 555-7823	3/17/2010	32	\$12.99	\$415.68	
9	Rhoden	Tony	24 Mountain Dr. Auburn, ME 04210	(207) 555-9915	3/24/2010	20	\$15.79	\$315.80	
10									
11									

Multiple Lines Within a Cell

- Click the cell in which you want to enter the text
- Type the first line of text
- For each additional line of text, press the Alt+Enter keys and then type the text

Figure 1-10

Two lines of text entered within a cell



Cell References

- Cell references are an important part of creating formulas in Excel.
- Using cell references allows your formulas to update automatically if the value in a particular cell changes and can also assist you in updating formulas as cells are copied or moved.
- There are two basic types of cell references, each serving a different purpose.
 - Relative Cell References
 - Absolute Cell References
 - Semi-Relative Cell References

Cell References

Relative Cell References

- References that will change in relation to the new location of the formula. Relative references identify cells based on their relationship to the cell containing the reference

Absolute Cell References

References that remain the same when a formula is copied to a new location. No matter where the formula or the values in the original cell are moved, the formula will continue to refer to the same cell

Semi-Relative Cell References

In Semi-Relative Cell References either the column remains unchanged or the row remains unchanged.

Cell References

D2 fx =C2*B2

	A	B	C	D
1	Student	Rate	Hours	Total
2	Bob	\$ 8.50	30	\$ 255.00
3	Anna	\$ 8.25	35	\$ 288.75
4	May	\$ 7.75	25	\$ 193.75

C11 fx =B11*\$C\$9

	A	B	C	D	E
9	Student Rate		\$ 8.50		
10	Student	Hours	Total		
11	Bob	30	\$ 255.00		
12	Anna	35	\$ 297.50		
13	May	25	\$ 212.50		

Function

- A function is a predefined formula that performs calculations using specific values in a particular order.
- In order to use functions correctly, you'll need to understand the different parts of a function and how to create arguments to calculate values and cell references.
- For example, to add the values in the range A1:A10, you could enter the following long formula:
`=A1+A2+A3+A4+A5+A6+A7+A8+A9+A10`
SUM function to accomplish the same thing:
`=SUM(A1:A10)`

Types of Excel function

- Excel has over 250 functions classified under the following categories:
- 1. Mathematical and trigonometric functions
- 2. Statistical functions
- 3. Financial functions
- 4. Look up and reference functions
- 5. Database functions
- 6. Logical functions
- 7. Information functions
- Date and Time functions

To use them click on Formulas bar you will get where to find those function

Inserting a function

- Click **Formulas** tab, Select **Insert function**
- From the **Insert function** dialog box that is displayed select a category of Function you want to use
- After selecting the category, different formulas of that category are displayed in select of function select box.
- Select a function from that select box.
- From the **function arguments** dialog box that is displayed provide different arguments as required.
- When you are finished, click OK.
- The function will show results in the active cell
- Or
- Type your function from the keyboard.

Basic function

Function	Example	Description
SUM	=SUM(A1:100)	finds the sum of cells A1 through A100
AVERAGE	=AVERAGE(B1:B10)	finds the average of cells B1 through B10
MAX	=MAX(C1:C100)	returns the highest number from cells C1 through C100
MIN	=MIN(D1:D100)	returns the lowest number from cells D1 through D100
SQRT	=SQRT(D10)	finds the square root of the value in cell D10
TODAY	=TODAY()	returns the current date (leave the parentheses empty)

IF function

- The IF function will check the logical condition of a statement and return one value if true and a different value if false. The syntax is
- **=IF (condition, value-if-true, value-if-false)**
- value returned may be either a number or text
- if value returned is text, it must be in quotes
- For example we could say
if the value of C4 is greater the 2000, calculate 10% of C4.,
otherwise calculate 20% of D4 . As a formula this would
be:
- **=IF(C4>2000,C4*0.10,D4*0.20)**

Comparison operators

Operator	Description
=	equals
>	is greater than
<	is less than
>=	is greater than or equal to
<=	is less than or equal to
<>	does not equal(i.e. is greater than or less than

Count and COUNTIF functions

- COUNT function is used when we want to count the number of cells in the range
- SYNTAX IS COUNT(RANGE)
- EXAMPLE : COUNT(R2:R30)

COUNTIF FUNCTION is used to count the number in the range with a condition

- Syntax COUNTIF(range, condition)
- Example: COUNTIF(B5:B70,">50")

Column Width and Row Height

- Drag the right border of the column heading left to decrease the column width or right to increase the column width
- Drag the bottom border of the row heading up to decrease the row height or down to increase the row height
- Select one or more columns or rows
- Click the Home tab on the Ribbon, click the Format button in the Cells group, and then click Column Width or Row Height
- Enter the column width or row height you want, and then click the OK button
- Or
- Double-click the line separating the columns or separating the rows

Inserting Columns, Rows, Cells

- Select the column(s), row(s) or cell(s) where you want to insert the new column(s) or row(s); Excel will insert the same number of columns or rows as you select
- In the Cells group on the Home tab, click the Insert button
- Or
- Right-click a column, row heading, selected column, row headings or selected cells, and then click Insert on the shortcut menu.

Deleting Columns, Rows, Cells

- Select the column(s), row(s) or cell(s) you want to delete; Excel will delete the same number of columns, rows or cells as you select
- In the Cells group on the Home tab, click the delete button
- Or
- Right-click column(s), row headings or selected cells, and then click Delete on the shortcut menu or hit delete key on the keyboard.

NB: You will have to confirm whether the cells below will shift up or whether the cells on the right will shift on the left.

Renaming a Worksheet

- To rename a worksheet, you double-click the sheet tab to select the sheet name, type a new name for the sheet, and then press the Enter key
- Sheet names cannot exceed 31 characters in length, including blank spaces
- The width of the sheet tab adjusts to the length of the name you enter
- **To access more paste options**
- To access more paste options, click the drop-down arrow on the Paste command.
- NB: It is easy to access commands quickly by right-clicking. Simply select the cell(s) you wish to format, then right-click the mouse.

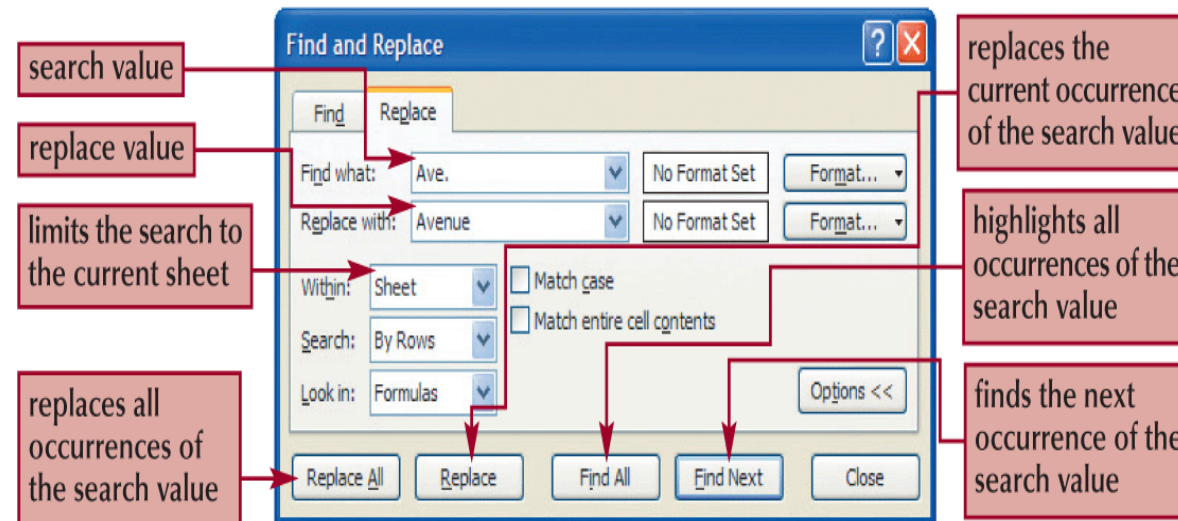
Inserting and Deleting a Worksheet

- To insert a new worksheet into the workbook, right-click a sheet tab, click Insert on the shortcut menu, select a sheet type, and then click the OK button
- You can delete a worksheet from a workbook in two ways:
 - You can right-click the sheet tab of the worksheet you want to delete, and then click Delete on the shortcut menu
 - You can also click the Delete button arrow in the Cells group on the Home tab, and then click Delete Sheet

Using Find and Replace

- You can use the **Find** command to locate numbers and text in the workbook and the **Replace** command to overwrite them

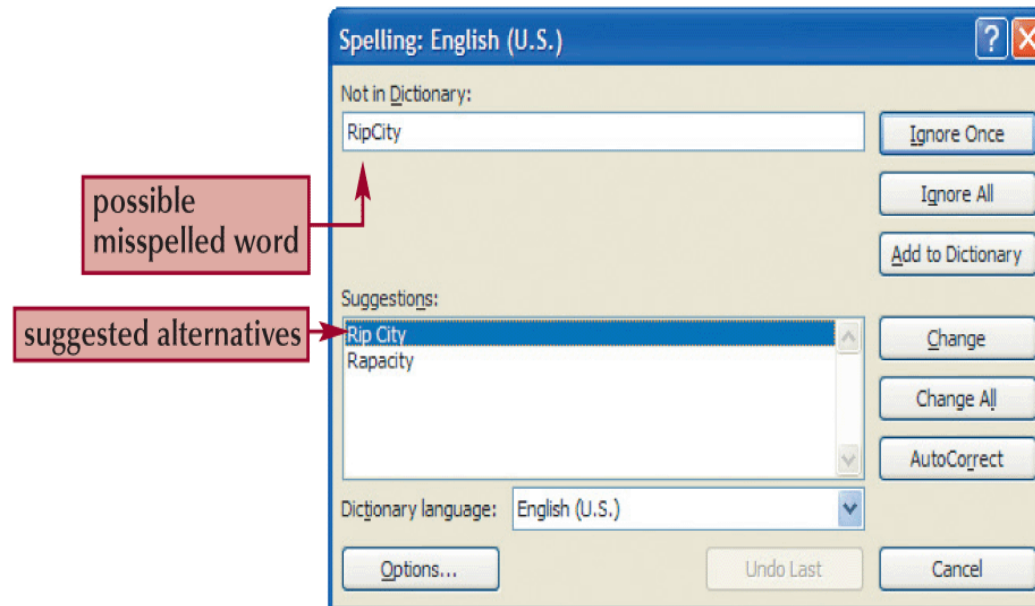
Figure 1-29 Find and Replace dialog box



Using the Spelling Checker

- The **spelling checker** verifies the words in the active worksheet against the program's dictionary

Figure 1-31 Spelling dialog box



Selecting Cell Ranges

To select an adjacent range:

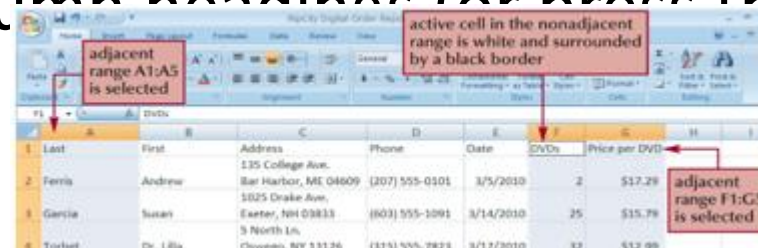
- Click the cell in the upper-left corner of the adjacent range, drag the pointer to the cell in the lower-right corner of the adjacent range, and then release the mouse button

To select a nonadjacent range of cells:

- Select a cell or an adjacent range, press the Ctrl key as you select each additional cell or adjacent range, and then release the Ctrl key

To select all the cells in a worksheet:

- Click the Select All button located at the intersection of the row and column headings (or press the Ctrl+A keys)



Moving or Copying a Cell or Range

- Select the cell or range you want to move or copy
- Move the mouse pointer over the border of the selection until the pointer changes shape
- To move the range, click the border and drag the selection to a new location (or, to copy the range, hold down the Ctrl key and drag the selection to a new location)

Inserting or Deleting a Cell Range

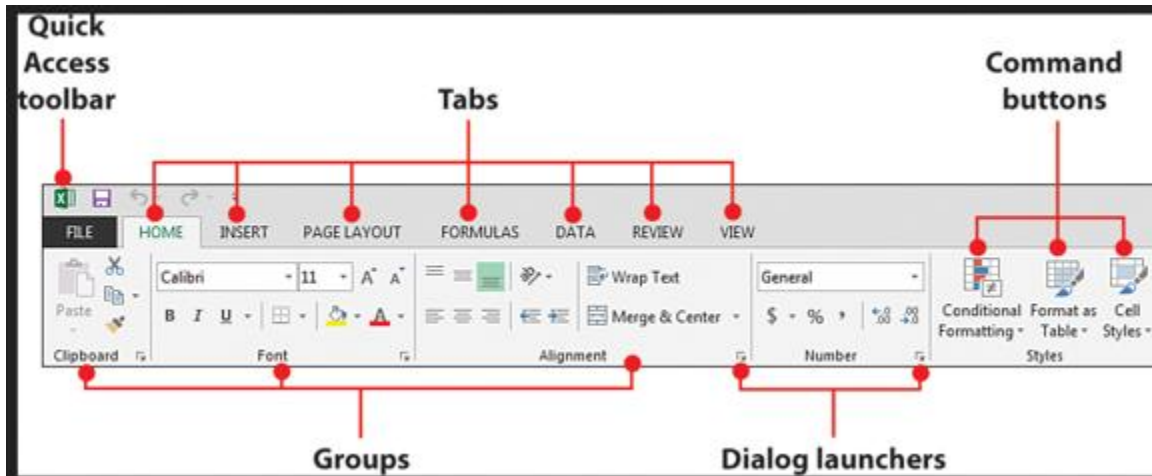
- Select a range that matches the range you want to insert or delete
- In the Cells group on the Home tab, click the Insert button or the Delete button

or

- Select the range that matches the range you want to insert or delete

Inserting or Deleting a Cell Range

- In the Cells group, click the Insert button arrow and then click the Insert Cells button or click the Delete button arrow and then click the Delete Cells command (or right-click the selected range, and then click Insert or Delete on the shortcut menu)
- Click the option button for the direction in which you want to shift the cells, columns, or rows
- Click the OK button



Creating charts

- Charts allow you to present data entered into the worksheet in a visual format using a variety of graph types.
- Choosing the right type of chart is just as important as deciding to use a chart at all. Using the best chart type and format will help you display your data visually in the most meaningful way

Inserting a chart

- Select the cells you want the chart to represent
- Click on Insert tab
- Select the type of chart you want to use
- Follow the instructions until the chart is created

Inserting chart wizard

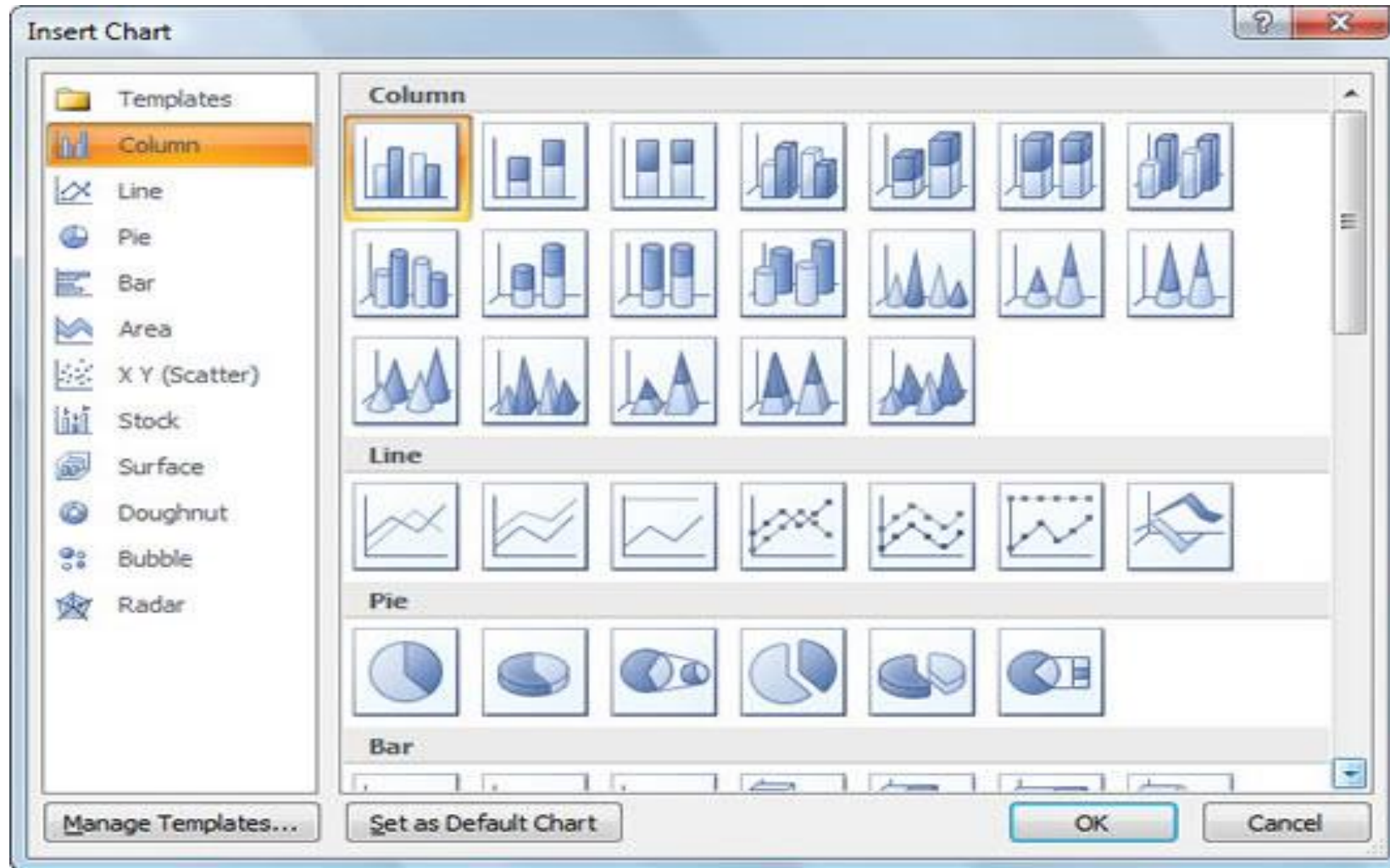


Chart types

- **Column charts:** The column chart, unlike the bar chart to which it is often compared, emphasizes variation over a period of time
- **Line charts:** A line chart shows the relationship of the changes in the data over a period of time.
- **Pie charts:** Pie charts contain just one chart data series. A pie chart shows the relationship of the parts to the whole
- **Bar charts:** A bar chart (horizontal bars) emphasizes the comparison between items at a fixed period of time

Chart types

- **Area charts:** An area chart shows the relative importance of values over time.
- **XY (Scatter) charts:** Scatter charts are useful for showing a correlation among the data points that may not be easy to see from data alone

Macros

- A macro is a series of actions in an Excel worksheet.

Recording a macro

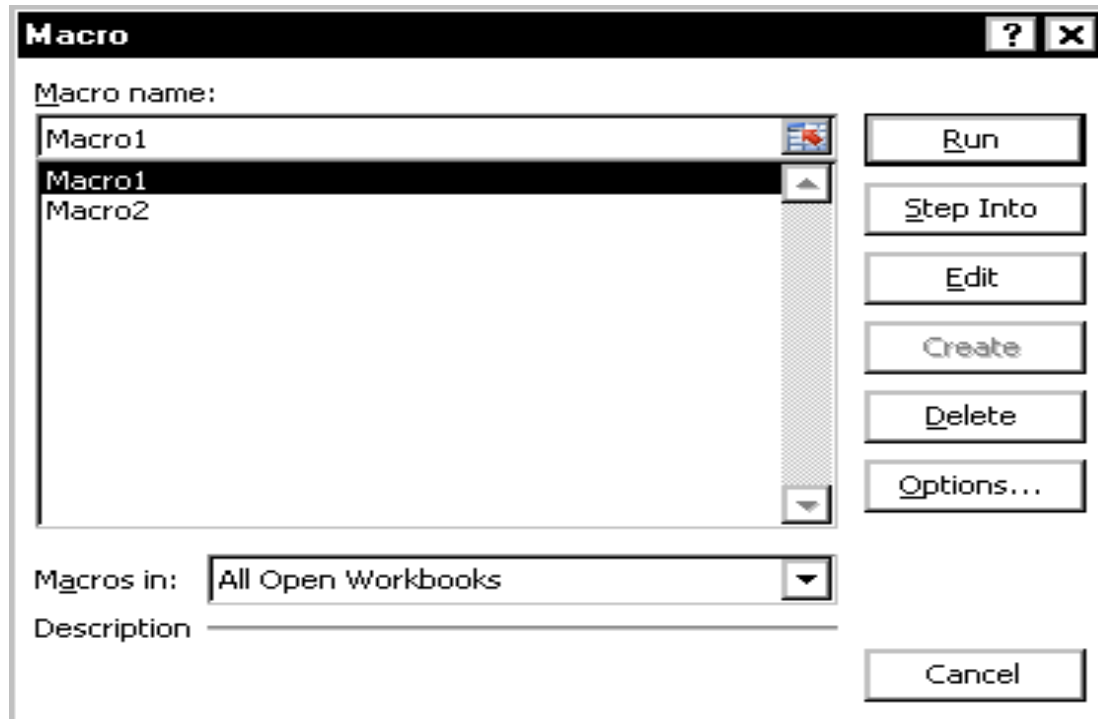
- On the **Developer** tab, in the **Code group**, click **Record Macro**. In the **Macro name** box, enter a name for the macro. Note that The first character of the macro name must be a letter. Subsequent characters can be letters, numbers, or underscore characters. Spaces cannot be used in a macro name.
- NB: macro name must not override any equivalent default Excel shortcut key.
- In the **Store macro in** list, select the **workbook** where you want to store the macro. If you want a macro to be available whenever you use Excel, select **Personal Macro Workbook**

Macros

- In the **Description** box, type a description of the macro. Click **OK** to start recording and perform the actions that you want to record. Once the actions to be recorded are completed, click **Stop Recording** on the **Developer** tab, in the **Code group**.
- **Running a Macro**
- You can run a macro only after it has been created and recorded.
- From the *Developer command tab, in the Code group, click MACROS* The Macro dialog box appears.
- From the *Macro name scroll box, select the macro you want to run*
- Click RUN.

Macros

- The *Macro dialog box closes and the selected macro performs the steps it recorded.*



Changing Worksheet Views

- You can view a worksheet in three ways:
 - **Normal view** simply shows the contents of the worksheet
 - **Page Layout view** shows how the worksheet will appear on the page or pages sent to the printer
 - **Page Break Preview** displays the location of the different page breaks within the worksheet

Working with Portrait and Landscape Orientation

- To change the page orientation:
 - Click the **Page Layout** tab on the Ribbon
 - In the Page Setup group, click the **Orientation** button, and then click **Landscape**
 - The page orientation switches to landscape

You can also print the contents of your workbook by using the Print command on the Office Button and choose Print after checking in **Page Layout view** that shows how the worksheet will appear on the page or pages sent to the printer

